

Agile Development Model

1. Introduction

The Agile development model is adopted for the **OmniSearch – Offline Multimodal RAG Intelligence Platform** to support incremental development, rapid experimentation, and continuous improvement of AI-driven components.

The system integrates multiple complex modules such as document ingestion, OCR processing, speech-to-text conversion, embedding generation, semantic search, vector indexing, and local LLM-based response generation. Since AI model behavior, retrieval accuracy, and user interface usability may require iterative refinement, Agile enables flexible and adaptive development.

Instead of building the entire multimodal RAG system in a single phase, the project is divided into structured sprints. Each sprint delivers a functional and testable module of the platform, allowing performance validation, accuracy evaluation, and optimization at every stage.

2. Why Agile is Suitable for This Project

The OmniSearch platform combines multiple AI and software components including:

- Document parsing (PDF, DOCX)
- OCR for images
- Speech-to-text processing
- Embedding generation
- Vector database indexing
- Semantic retrieval
- Local LLM-based answer generation
- Citation linking and source navigation

Since retrieval quality, embedding performance, and LLM responses may require tuning and optimization, Agile is well-suited for this project because it allows:

- Incremental development of multimodal ingestion pipelines
- Early testing of embedding accuracy and retrieval relevance
- Iterative improvement of search ranking algorithms

- Progressive integration of AI components
- Continuous refinement of user interface and response formatting
- Rapid debugging of model or indexing issues

Agile ensures that core functionalities such as semantic retrieval and grounded answer generation are implemented early and enhanced iteratively.

3. Agile Workflow

The development follows repeated Agile cycles consisting of:

1. Requirement Planning

Features are selected from the Software Requirements Specification (SRS), such as:

- Document ingestion
- OCR processing
- Speech-to-text conversion
- Embedding generation
- Semantic search
- Answer generation with citations

Each sprint focuses on a specific subsystem.

2. System Design

Architecture design is prepared for selected modules including:

- Class design
- Data flow diagrams
- Indexing structure
- Retrieval pipeline
- Model interaction flow

Design ensures modularity and offline compatibility.

3. Development

Selected features are implemented using:

- Local AI models
- Vector database
- Backend logic
- User interface components

Each feature is implemented independently but designed for integration.

4. Testing

Each module is tested using:

- Sample PDFs, images, and audio files
 - Query test cases
 - Accuracy evaluation of retrieved results
 - Verification of citation correctness
 - Performance testing for indexing and search speed
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5. Review and Optimization

The module is reviewed and optimized based on:

- Retrieval accuracy
- LLM hallucination reduction
- Processing speed
- Memory usage
- User experience

Improvements are incorporated before the next sprint.

6. Next Sprint

The next set of features is planned and developed, gradually building a complete offline multimodal RAG system.

4. Sprint Plan for This Project

Sprint 1 – Core Document Ingestion

- Upload PDF and DOCX files
 - Text extraction from documents
 - File validation and metadata storage
 - Basic file indexing structure
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Sprint 2 – Image and Audio Processing

- OCR for images and screenshots
 - Speech-to-text conversion for audio
 - Transcript generation
 - Integration of multimodal text extraction
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Sprint 3 – Embedding & Vector Indexing

- Embedding generation for extracted text
 - Chunking of long documents
 - Vector database integration
 - Similarity search implementation
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Sprint 4 – Semantic Retrieval Engine

- Query embedding generation
 - Top-k retrieval logic
 - Ranking and filtering mechanisms
 - Retrieval performance optimization
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Sprint 5 – LLM Integration & Grounded Answering

- Local LLM integration
 - Context-based answer generation
 - Citation attachment mechanism
 - Hallucination filtering
 - Answer formatting
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Sprint 6 – UI, Optimization & Security

- Unified chat-based interface
 - Source preview and citation navigation
 - Performance tuning
 - Memory optimization
 - Offline testing validation
 - Bug fixing and final refinements
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5. Advantages of Using Agile

The Agile model enables the OmniSearch platform to be developed in a structured yet flexible manner. By implementing AI modules incrementally, issues related to embedding quality, retrieval accuracy, or LLM responses are identified early and corrected without redesigning the entire system.

Agile supports:

- Continuous improvement of AI model performance
- Iterative refinement of retrieval algorithms
- Progressive integration of multimodal capabilities
- Early detection of integration issues
- Efficient optimization of offline execution

This results in a more accurate, reliable, and scalable multimodal intelligence platform.

6. Conclusion

By following the Agile development model, the OmniSearch – Offline Multimodal RAG Intelligence Platform can be developed iteratively and efficiently. Each sprint delivers measurable progress in ingestion, indexing, retrieval, and generation capabilities.

Continuous testing, refinement, and optimization ensure that the final system provides accurate, grounded, and transparent responses across multiple data formats while operating completely offline.